Yufeng (Alex) Xia

Email: yufeng.xia@ed.ac.uk
xyf2002.github.io linkedin.com/in/yufeng-xia-12648a25b
Mobile: (+44) 7918155093
Edinburgh, UK

Research Interests

- I work on **how high-level computation is turned into efficient parallel execution, and on what that execution costs on real hardware**. This has meant two things. First, **compiler-driven transformation**: taking a declarative or high-level specification and generating vectorised, quantised or otherwise specialised code automatically, rather than asking a programmer to hand-tune it. Second, **parallel execution under contention**: deciding how a computation is partitioned across accelerators, and what happens to it when that capacity is taken away mid-execution. I am drawn to problems where the two meet, and would like to bring this experience to the automatic vectorisation and parallelisation of declarative computations and to the systems that execute them at scale.

Education

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
  ICSA; supervised by Prof. Mahesh K. Marina *Sep 2025 – Nov 2026 (expected)*
  - **Thesis:** *Flux-DT: An Elastic, Preemption-Tolerant Runtime for Stateful Workloads on Spare Compute.*
- **University of Edinburgh — BSc Computer Science** Edinburgh, UK
  *Grade: **First Class Honours*** *Sep 2021 – May 2025*
  - **Relevant coursework:** Machine Learning Systems (**A**), Computer Architecture and Design (**A**), Computer Systems (**A**).

Research Experience

Compilers and Parallel Execution

- **CoSenseAQ — Automatic Quantisation in LLVM from Sensor Specifications** under submission
  *Junior Research Assistant, ICSA; **first author**; supervised by Dr Antonio Barbalace* *Apr 2024 – May 2025*
  - **Transformation derived from a specification, not from source**: Built an LLVM framework that derives a quantisation strategy from sensor technical specifications — information a compiler cannot recover from the program itself — and applies it as IR-level transformations, replacing manual per-target tuning. Implemented as LLVM passes in C++ over a cross-compilation toolchain.
  - **Correctness and measured cost**: Established that the transformation stays within the numerical behaviour the specification permits; measured on ARM Cortex-M0 and Cortex-A53 in code size, execution time, power and area.
- **GPU Kernel Design for Similarity Search** grade **A**
  Machine Learning Systems coursework, University of Edinburgh — individual project *2024*
  - **Vectorised kernels written against the memory hierarchy**: Wrote Triton kernels for L2, cosine, dot-product and Manhattan distance, and built KNN, K-Means and approximate nearest-neighbour search on them. Performance came from explicit control of the mapping: block-level tiling of the reduction dimension, masked handling of ragged boundaries, and coalesced loads.
- **RISC-V Processor Design and FPGA Implementation** grade **A**
  Computer Architecture and Design coursework, University of Edinburgh — individual project *2023*
  - **From RTL to measured cycles**: Implemented the ALU, register file and datapath in Verilog, integrated them into a working RISC-V processor, and validated it on a PYNQ-Z2 board after synthesis in Vivado; designed microarchitectural enhancements and evaluated them in CPI and memory bandwidth.

Systems for Machine Learning

- **Flux-DT — Preemption-Tolerant Runtime for Stateful Workloads** **MSc thesis**
  University of Edinburgh — sole author *Sep 2025 – Present*
  - **Partitioning and shedding as one decision**: Compute-sharing designs assume the harvesting workload is stateless and disposable; this one is neither, so reclaiming its compute cannot mean killing it. Formulated work placement as a joint objective across mutually interfering units and designed a greedy heuristic deciding which work is worth doing; the same ranking fixes the shedding order when compute is reclaimed mid-solve. A control loop answers resource pressure with graded actions — relocation, fidelity reduction, suspend and resume — rather than a single eviction.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026, **Best Paper**
  Research Assistant — C++ backend, fault tolerance, GPU partitioning *Oct 2025 – Mar 2026*
  - **Per-GEMM SM partitioning**: Implemented trace-driven GPU capacity partitioning with CUDA green contexts, dispatched through autograd hooks at per-GEMM granularity, so co-located training workloads share a device under explicit rather than incidental allocation.

- **Event-driven backend and failure handling**: Built the proxy server, client and device-tracking layer in C++ on an epoll event loop with Protobuf messaging, supporting up to 1,800 emulated devices on one GPU server; added failure detection and target reselection, partition re-derivation, and runtime membership changes, removing a class of failure-induced deadlocks.

- **Weaver — Foundation Model Training over Shared AI Compute** under submission
  Research Assistant — controller evaluation, profiling, large-scale runs *2024 – 2026*
  - **Harvesting accelerator capacity without disturbing a co-tenant**: Evaluated the controller deciding how much accelerator capacity a training workload may harvest from infrastructure shared with a latency-critical tenant. On a multi-site testbed it harvests up to **83%** of the available spare compute with no degradation to the co-tenant; characterised their interference through GPU profiling and ran the large-scale distributed training used in the evaluation.

- **EmuRAN — Scalable and Cost-Effective Cloud-based RAN Emulation** MobiCom 2026, cond. accepted
  *Research Assistant — **owned the end-to-end evaluation*** *2024 – 2026*
  - **Kernel-level virtualisation, evaluated at scale**: Built and deployed a custom Linux kernel module and a modified KVM module providing time-dilated virtual machines, with automated provisioning, mesh routing and Kubernetes formation; ran the system across **130 cloud instances and 6,400 CPU cores**, emulating over **10,000** devices, and designed the fidelity and scalability evaluation.

Publications

EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System *MobiCom'26 (cond. accepted)*
*Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*

Morphling: Emulator for Distributed Machine Learning at the Edge *EdgeSys'26 (Best Paper)*
*Leyang Xue, **Yufeng Xia**, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina*

Wasp: Foundation Model Training System for the Edge *SEC'26*
*Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina*

CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization *under submission*
***Yufeng Xia**, Pei Mu, Antonio Barbalace*

Weaver: Foundation Model Training over AI-RAN Compute Infrastructure *under submission*
*Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*

Technical Skills

- **Languages** C, C++ (primary), Python, Verilog, Bash, Java.
- **Compilers & Kernels** LLVM pass development, IR-level analysis and transformation, automatic quantisation, cross-compilation toolchains; Triton kernel development (tiling, masking, memory coalescing), CUDA green contexts and SM partitioning, GPU profiling.
- **Systems** Distributed runtimes in C++ (epoll event loops, Protobuf messaging), scheduling under contention and preemption, Linux kernel modules and custom kernel builds, KVM virtualisation, multi-node cloud deployment (Azure, GCP, CloudLab); Docker, Kubernetes, Git.
- **Architecture & ML** RTL design in Verilog, FPGA synthesis and on-board validation (Vivado, PYNQ-Z2), quantitative evaluation of CPI, cache behaviour and memory bandwidth; PyTorch, distributed foundation-model training, evaluation design.

Awards, Industry Experience and Service

- **Awards Best Paper**, EdgeSys 2026 (co-located with ACM MobiSys 2026), for *Morphling*; **First Class Honours**, BSc Computer Science.
- **Industry** Back-end Development Intern, Senior Tech, Suzhou, China (Jun–Sep 2023): production backend APIs in a commercial engineering team, live platform migration to Spring Boot, code review and release testing (Java, MySQL, Redis, Kafka).
- **Service & Languages** Student volunteer, EuroSys 2026 (on-site conference operations); Mandarin (native), English (fluent; IELTS 7.0, GRE 336).

References

- **Prof. Mahesh K. Marina** MSc by Research supervisor — School of Informatics, University of Edinburgh • mahesh@ed.ac.uk
- **Dr Antonio Barbalace** BSc dissertation supervisor and Junior Research Assistant supervisor — School of Informatics, University of Edinburgh • antonio.barbalace@ed.ac.uk